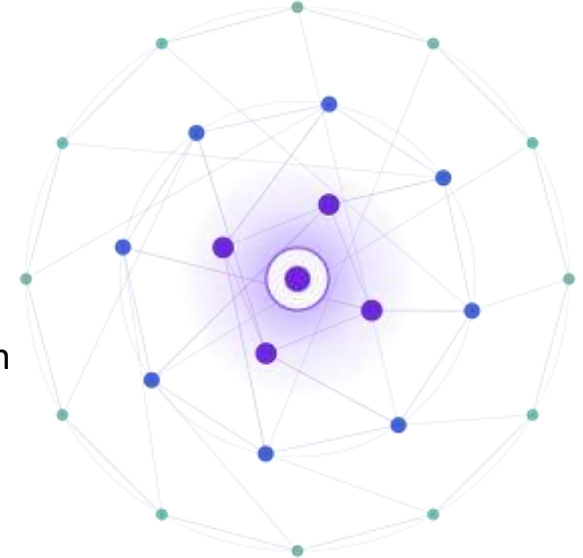


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 2
- Team Name - InnoVerse
- College Name - SRM Institute of Science and Technology
- Member Name & Email 1- Swarnava Sarkar & swarnava2019@gmail.com
- Member Name & Email 2 – K. Trimman Choudhary & kk0569@srmist.edu.in
- Member Name & Email 3- Utsav Singh & saxenautsav06@gmail.com
- Member Name & Email 4- Hemanth Muthyalu & mr2853@srmist.edu.in
- Submission Github link - <https://github.com/SWARNAVA182006/samsung-guided-troubleshooting.git>

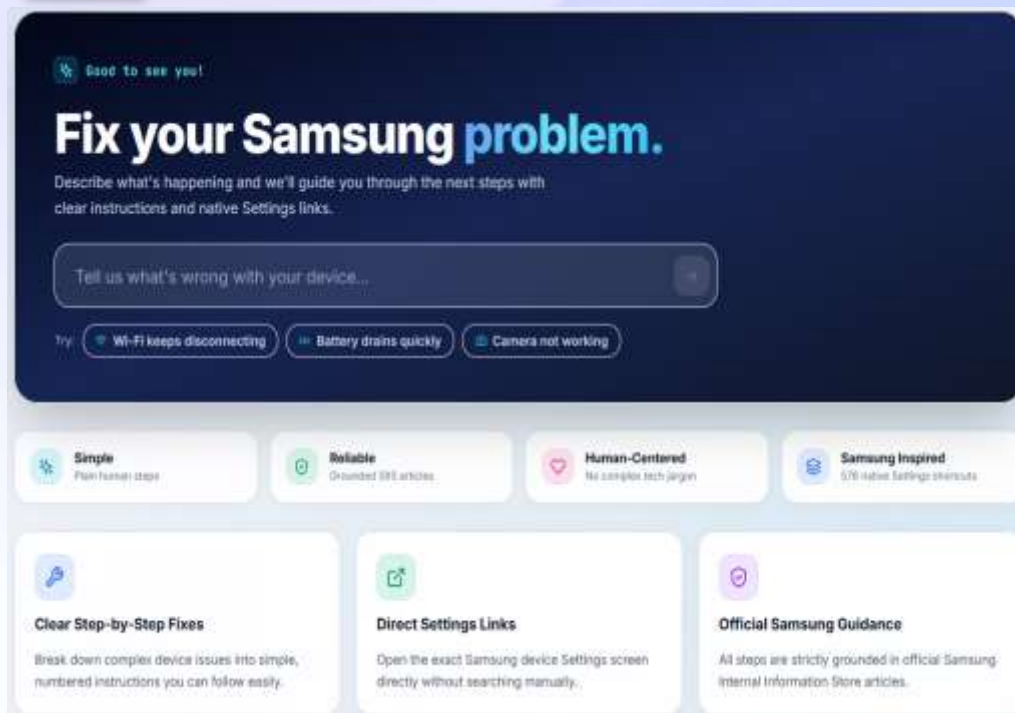


Samsung Guided Troubleshooting Engine

Turn vague device complaints into grounded, actionable troubleshooting.

Support knowledge exists. The friction is turning it into action.

Users speak naturally; support systems still make them translate the problem.



TODAY

01

Read SIIS

Find the relevant article

02

Interpret

Translate guidance into steps

03

Navigate

Find the right Settings screen

04

Explain

Tell the user what to tap

THE MISSING LAYER → guided, structured, actionable output

Existing Solutions & Gaps

Generic AI talks. Rules break. We built for the contract.

The difference is not another chatbot — it is the verified path from language to action.

GENERIC LLM

- ✗ Free-form response
- ✗ Weak structural guarantees
- ✗ No catalog-backed deeplink contract

RULE-BASED

- ✗ Brittle on paraphrases
- ✗ Manual maintenance
- ✗ Limited generalization

OUR ENGINE

- ✓ Official SIIS grounding
- ✓ Validated structure + deeplinks
- ✓ Guided user experience

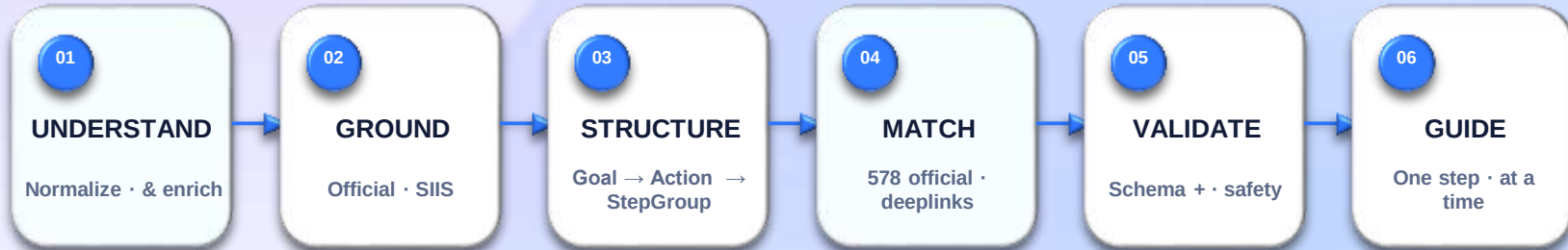
Contract **protected** in code.

KEY INSIGHT Troubleshooting is not just a conversation — it is a verifiable action plan.

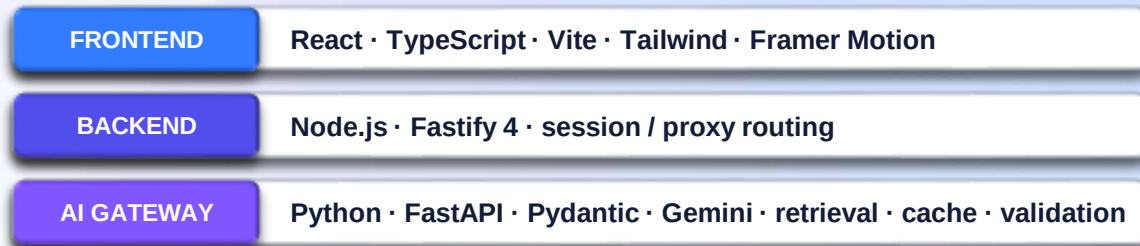
Our Solutions & Architecture Diagram

From natural language to a verified Settings action.

An end-to-end pipeline with grounding, validation and fast-path caching.



IMPLEMENTATION



OFFICIAL CONTRACT

Complaint + SIIS

Goal → Action → StepGroup
+ catalog-backed deeplink

FAST PATH · Exact + paraphrase cache

Demo & Product Walkthrough

The output is a guided journey — not a paragraph dump.

A user describes the problem once; the product carries the complexity.

SAMSUNG BENCHMARK CASES (20) — For judges & evaluation

[row_1] row_1: Email server not responding on Samsu... Connectivity	[row_2] row_2: Blank or black display on a Samsung ... Display	[row_3] row_3: Some things to check first... Display	[row_4] row_4: Blank or black display on a Samsung ... Display	[row_5] row_5: Transfer Secure folder with Smart Sw... Data & Backup
[row_7] row_7: Use Multi window and App pairs on yo... Display	[row_8] row_8: Screen mirroring to your Samsung TV... Connectivity	[row_9] row_9: Access your Galaxy phone's data if I... Display	[row_10] row_10: Screen flickers when using the Camer... Camera	[row_11] row_11: Some things to check first... Display

Describe Device Problem Guided Engine

DEVICE PROBLEM / SYMPTOM *

Describe what's wrong with your Samsung device (e.g. My screen keeps flickering whenever I open Gmail)...

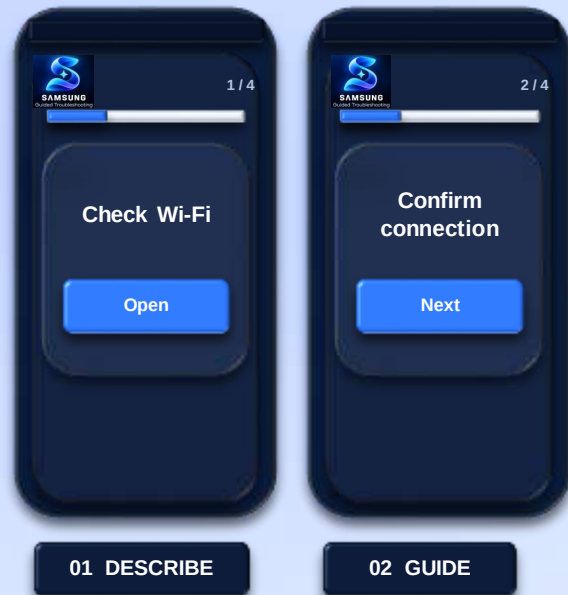
Grounding Knowledge Article (SIS)

Start Guided Fix

Ready to Fix Your Device

Describe your device problem and get grounded troubleshooting steps with official Samsung Settings shortcuts.

Grounded Fixes 678 Settings Links



Tools and tech stack used

Purpose-built layers, not buzzwords.

Every technology shown here is tied to an actual responsibility in the system.

PRODUCT

React + TypeScript + Vite + Tailwind CSS + Framer Motion

Premium responsive UI, guided state, themes and motion

API

Node.js + Fastify

Session routing and service boundary

AI GATEWAY

Python + FastAPI + Pydantic + Gemini (optional)

Generation, grounding, validation and safe output

INTELLIGENCE

SIIS grounding + TF-IDF retrieval + deeplink matching + cache

Deterministic retrieval, catalog integrity and fast-path reuse

DESIGNED AROUND THE CONTRACT

Grounded

Structured

Catalog-backed

Validated

Impact & Use case

Reduce the distance between “I have a problem” and “I know what to do.”

The same underlying engine serves users, support workflows and SIIS automation.

20

OFFICIAL SIIS CASES

578

OFFICIAL DEEPLINK
ENTRIES

0

URL LEAKS IN
EVALUATED OUTPUT

BEFORE

Read → Interpret → Search → Explain → Navigate

AFTER

Describe → Ground → Guide → Open → Confirm



END USERS

Clear steps, less Settings hunting



SUPPORT

Structured, repeatable guidance



SIIS PIPELINE

New articles can flow through
the same API

Innovation highlights, results and limitations

The model generates. The system verifies.

Measured benchmark facts are kept separate from engineering limitations.

GROUNDING

Steps derive from supplied SIIS guidance.

ACTIONABLE

Deeplinks come from the official catalog.

DETERMINISTIC

Contract checks run after generation.

REPRODUCIBLE

Evaluation is scriptable and inspectable.

HONEST BOUNDARIES

Candidate retrieval rate $\neq$ semantic accuracy • Jaccard cache has limits

BENCHMARK SNAPSHOT

20 / 20

Schema-valid outputs

0

URL leaks

146

Catalog-backed
deeplinks

1.68 s

Average 20-case
pipeline*

**Average local pipeline latency across the 20 official cases; not a p95 claim.*

What's next

Build from a solid contract toward deeper device-aware assistance.

Current system first; future improvements are clearly separated.

CURRENT

Grounded Theme 02 contract

Validated output
Catalog-backed actions

01

NEXT

Ground-truth relevance set

Stronger retrieval
evaluation
Semantic cache

02

FUTURE

Multi-turn troubleshooting

Device-aware guidance
Native Galaxy integration

03

Scale the intelligence, keep the contract.

More semantic depth without losing verification.

Brownie points slide(differentiation)

From language to action — with verification at every layer.

A consumer-facing experience built on a deterministic engineering core.

01

GROUNDING FIRST

Official SIIS is the source of truth.

02

CONTRACT, NOT PROMPT HOPING

Programmatic validation protects the output.

03

TEXT → ACTION

Catalog-backed Settings deeplinks make guidance actionable.

04

REAL PRODUCT UX

Guided steps, completion, history and responsive design.

AI

VERIFIED
ACTION

THE ENGINE

20 SIIS cases

578 deeplink entries

11 validation rules

0 URL leaks

**Built to be
inspectable.**

Checklist- Updated on Public GitHub

Everything important is visible, reproducible and reviewable.

Keep this slide quiet; the proof is in the checked boxes.



WORKING PROTOTYPE

Guided troubleshooting UI + API



TEST SUITE

pytest + Jest + frontend build



PUBLIC GITHUB

github.com/SWARNAVA182006/samsung-guided-troubleshooting



DEMO VIDEO

https://drive.google.com/file/d/1TwWykxGNnVJEitRnafxDy2_izLY_SLDM/view?usp=sharing



REPRODUCIBLE SETUP

README + Docker + environment configuration



AI DISCLOSURE

Required submission document



EVALUATION OUTPUT

20 official benchmark cases + reports



PRESENTATION

PPTX + PDF final deliverable

FINAL CHECK • FACT-CHECKED • RENDERED • READY FOR REVIEW



Samsung Guided Troubleshooting Engine

From vague complaints
to **clear, actionable fixes.**

20/20 official cases
evaluated

578 official deeplink
entries

0 URL leaks

InnoVerse • SRMIST, Kattankulathur

<https://github.com/SWARNAVA182006/samsung-guided-troubleshooting.git>

Thank You!

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India